

SCADA Security

SCADA security is the practice of protecting SCADA networks from any security attack or threat. These networks, just like any other network, are under threat from cyber-attacks that could bring down any part of the nation's critical infrastructure quickly and with dire consequences if the right security is not in place. Capital expenditure is another key concern; SCADA systems can cost an organization from tens of thousands to millions of shillings. For these reasons, it is essential that organizations implement robust SCADA security measures to protect their infrastructure and the millions of people that would be affected by disruption caused by an external attack or internal error.

SCADA Network Security Threats

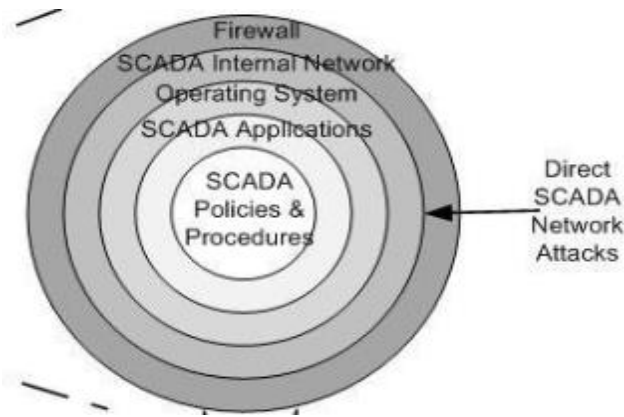
- *Hackers:* Individuals or groups with malicious intent could bring a SCADA network to its knees. By gaining access to key SCADA components, hackers could unleash chaos on an organization that can range from a disruption in services to cyber warfare.
- *Malware:* Malware, including viruses, spyware and ransomware can pose a risk to SCADA systems. While malware may not be able to specifically target the network itself, it can still pose a threat to the key infrastructure that helps to manage the SCADA network. This includes mobile SCADA applications that are used to monitor and manage SCADA systems.
- *Terrorists:* While hackers are usually motivated by sordid gain, terrorists are driven by the desire to cause as much mayhem and damage as possible.
- *Employees:* Insider threats can be just as damaging as external threats. From human error to a disgruntled employee or contractor, it is essential that SCADA security addresses these risks.

Risks a SCADA system could suffer after cyber attack

- i. Deletion of system files from the server
- ii. Plant a virus and take complete control of the system
- iii. Modify any logged data in the database system
- iv. Log any company-sensitive data for personal or competitive usage
- v. Log keystrokes from operators and obtain usernames and passwords

SCADA Ring of defense

The figure shows the “rings of defense” of the SCADA networks. Attacks on the SCADA system can be external, via the Internet through the corporate network, or internal which can influence from the corporate network or the SCADA system from the RTU level or the application level. The development of a corresponding security strategy includes an analysis of multiple layers of the corporate network and the SCADA system architecture. This includes firewalls, internal networks, operating systems, applications, communications, policies and security procedures)



SCADA maintenance

Maintaining the various components that comprise a Supervisory Controls and Data Acquisition (SCADA) system isn't always easy. Below are some of the tips to operate a smooth SCADA network

- Use standardized systems for alarm identification: critical alarms should be easily distinguished from warning alarms and other distractions
- Adopt systems that drive consistency between the various locations to ensure efficiency and safety.
- Ensure secured internal and external data backup systems which can easily be accessed in time of need
- Prevent unauthorized entry to the system to improve the network security without comprising efficiency
- Use integrated data storage so that data can be shared among authorized users. Manual transfer process is time wasting and error prone.
- Provide training to employees to alert them on the new technologies, threats and necessary protection measures needed to protect SCADA new network environment.
- Perform regular security testing: of both infrastructure control systems and RTUs, of SCADA networks needs to evolve from a technical basis to a more risk based, process-oriented management basis.
- Replace obsolete components

Examples of standardization companies for SCADA systems

- Institute of Electrical and Electronic Engineers (IEEE)
- American National Standards Institute (ANSI)
- International Electrotechnical Commission (IEC)
- Distributed Network Protocol (DNP) users

Applications of SCADA

SCADA has made comprehensive use of features such as flexibility, reliability, and scalability in automating complex systems. There are countless applications in the real world where

SCADA has already been effective in providing surveillance and control alternatives across a broad spectrum of sectors, from energy production to agricultural systems. SCADA is widely used in different areas from chemical, gas, water, communications and power systems

- **Electric Power Generation, Transmission, and Distribution:** Using SCADA systems, electrical utilities detect current flow and line voltage, monitor circuit breaker operation, and take sections of the power grid online or offline.
- **Mass transit and Railway Traction:** Transit officials use SCADA to regulate electricity for subways, trams and trolley busses; to automate railway traffic signals; to monitor and identify trains and busses, and to control railway crossing doors
- **Manufacturing Units:** SCADA systems are used to regulate industrial automation and robots, and monitor process and quality control
- **Water, Waste Water Utilities and Sewage:** Municipal water utilities use SCADA to monitor and control water flow, tank concentrations, pipe pressure, and other variables
- **Buildings, Facilities, and Environments:** Facility managers use SCADA to regulate HVAC, cooling, lighting and input systems.
- **Thermal Power Plants:** Most operational inspections of thermal power plants are automatic. However, manual action may be necessary at times. The plant is therefore equipped with monitoring and alarm systems that alert plant operators when certain working parameters deviate significantly from their normal range. The demand for greater reliability and effectiveness is growing in thermal power plants. After periodic intervals, the power plant needs continuous inspection and tracking. There may be chances of mistakes when human employees measure at different phases. To improve reliability, automation is required to improve the general power plant efficiency. Automation is established through the use of PLC & SCADA, which decreases human workers' mistakes. SCADA system is used to supervise a complete process.
- **Forestry, Pulp and Paper Industry:** The industry has ample uses of SCADA ranging from automation & process control, energy management, drive control, power protection, enclosure systems, and safety. SCADA systems are used throughout the paper supply chain – including within the wood yard, to the chippers, evaporators, digestors, refiners, cleaners, drying & pressing, and the paper machines. This complex processing of trees to paper is automated from end-to-end through the use of established SCADA systems.

PLC vs. SCADA

The differences between PLC and SCADA are listed below:

PLC	SCADA
It stands for Programmable Logic Controller.	It stands for Supervisory Control and Data Acquisition.
It comprises CPU, I/O modules, and the peripheral devices.	It comprises of computers, GUI (Graphical User Interface), and networked data communications.
It is hardware .	It is software .
It is a system that controls motors and complex machines.	It is a system to run and monitor the plant processes.
It is a hardware device that can be of various types, as per the requirements.	It consists of several components that increase its capability.

Examples of SCADA software

- ✓ Citect SCADA from Schneider Electric
- ✓ InTouch from Wonderware
- ✓ Experion SCADA from Honeywell
- ✓ iFix from General Electric
- ✓ Ignition from Inductive Automation
- ✓ Simatic WinCC V7 from Siemens